

Exercise Sheet for Maths Material Covered in Week 10

Vector Spaces and Matrices

Note that you will be given **feedback** for question (2).

(1) Throughout this question, the field we consider is the set of rational numbers.

- (a) Show that the two vectors $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ are not linearly independent, i.e., are linearly dependent, in the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers.
- Show that the four vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 14 \\ 5 \\ \frac{1}{2} \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ do not form a basis for the vector space $\mathbb{Q}^3$ of three-tuples of rational numbers. Show also that these four vectors are a spanning set.

Comment: Recall the definitions of linear independence and spanning set.

(2) (**feedback**) Throughout this question, the field we consider is the set of rational numbers.

- (a) Show that the two vectors $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ form a basis for the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers. Is this basis orthogonal?
- (b) Show that the three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ form a basis for the vector space $\mathbb{Q}^3$ of three-tuples of rational numbers. Is this basis orthogonal?

Comment: Recall that for a set of vectors to be a basis you need to show two conditions: they are linearly independent, and they are a spanning set, i.e., every vector can be spanned by them. For a basis to be orthogonal, you need to check that inner product of every pair of distinct vectors in the basis is 0.

(3) This question is about matrix multiplication. Let $A = \begin{pmatrix} 1 & 5 & 8 \\ 0 & 6 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 18 & 7 \\ 11 & -4 \end{pmatrix}$.

- (a) Can you multiply A with B to obtain the matrix AB ? If yes, find the matrix AB and show your steps? If not, explain why.
- (b) Can you multiply B with A to obtain the matrix BA ? If yes, find the matrix BA and show your steps? If not, explain why.

(4) Let $a, b, c, d \in \mathbb{Q}$ be such that $(ad-bc) \neq 0$. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $Y = \begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$.

Show that $XY = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = YX$.

Comment: Note that you are asked to compute two matrix multiplications in this question.

Question (5) is on the next page

(5) Verify that the set of all (2×2) matrices $\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Q} \right\}$ is a vector space over the field $\mathbb{Q}$ (which has the two operations $+$ and $\times$) with the following definition of vector addition & multiplication of a scalar with a vector:

(a) Vector addition: If $\vec{v}_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ then $\vec{v}_1 \oplus \vec{v}_2$ is defined to be

$$\begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{pmatrix}$$

(b) Multiplication of a scalar and a vector: If $\vec{v} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $r \in \mathbb{Q}$ then $r\vec{v} = \begin{pmatrix} r \times a & r \times b \\ r \times c & r \times d \end{pmatrix}$

Comment: Each of the 8 conditions in the definition of vector space over a field needs to be verified. You can use any of the conditions available due to $\mathbb{Q}$ being a field.